SOMNATH BASU ROY CHOWDHURY

Residential Information
Mountain View, CA
Mountain View 94043, US.

Electronic Information
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Education

University of North Carolina (UNC) at Chapel Hill Jan 2021 - present
Ph.D. in Computer Science
Advisor: Prof. Snigdha Chaturvedi
Thesis Committee: M. Bansal, G. Bertasius, H. He, A. McCallum

Indian Institute of Technology (IIT) - Kharagpur, India 2013 - 2018
B.Tech. & M.Tech. (Dual Degree Hons.) in Electrical Engineering Rank - 3/31
Minor in Computer Science & Engineering (CSE)
Micro-specialization in Embedded Wireless Systems (EWS)
Advisor: Prof. Jayanta Mukhopadhyay

Employment History

Google Research, Mountain View May 2025 - Present
Research Scientist, Algorithms & Optimization team
Manager: Amr Ahmed

Adobe Research, College Park May 2024 - Aug 2024
Research Intern, Document Intelligence Lab
Collaborators: A. Nenkova, V. Morariu

Google Research, Mountain View May 2023 - Aug 2023
Student Researcher, Algorithms & Optimization team
Collaborators: A. Dubey, N. Monath, A. Beirami, R. Kidambi

Google Research, Mountain View May 2022 - Dec 2022
Student Researcher, Algorithms & Optimization team
Collaborators: A. Dubey, K. Choromanski, N. Monath

Microsoft Corp., India Jul 2019 - Dec 2020
Data Scientist II, Satori Knowledge Graph (Bing team)

Apple Inc., India Jun 2018 - Jul 2019
Software Engineer, Identity Management Services (IdMS).

Publications

Google Scholar: xGbyrIUAAAAJ.

Conference & Journal Papers:

[27] **EUGens: Efficient, Unified, and General Dense Layers**
S. Min Kim*, B. Kim*, A. Sehanobish*, S. Basu Roy Chowdhury*, R. Kidambi*, D. Shim, A. Dubey, S. Chaturvedi, M. Oh, K. Choromanski
Proceedings of NeurIPS 2025. [pdf]

[26] **Fundamental Limits of Perfect Concept Erasure**
S. Basu Roy Chowdhury, A. Dubey, A. Beirami, R. Kidambi, N. Monath, A. Ahmed, S. Chaturvedi
Proceedings of AISTATS 2025. [pdf]

[25] **Towards Scalable Exact Machine Unlearning Using Parameter Efficient Fine-tuning**
S. Basu Roy Chowdhury, K. Choromanski, A. Sehanobish, A. Dubey, S. Chaturvedi
Proceedings of ICLR 2025. (*Preliminary version* at Safe GenAI Workshop, NeurIPS 2024) [pdf]

[24] **Exploring Safety-Utility Trade-Offs in Personalized Language Models**
A. Rao Vijjini*, S. Basu Roy Chowdhury*, S. Chaturvedi
Proceedings of NAACL 2025 (*Oral*). [pdf]

[23] **Structured Unrestricted-Rank Matrices for Parameter Efficient Finetuning**
A. Sehanobish*, A. Dubey*, K. Choromanski*, S. Basu Roy Chowdhury*, D. Jain, V. Sindhwani, S. Chaturvedi
Proceedings of NeurIPS 2024. [pdf]

[22] **Fast Tree-Field Integrators: From Low Displacement Rank to Topological Transformers**
K. Choromanski*, A. Sehanobish*, S. Basu Roy Chowdhury*, H. Lin*, A. Dubey, T. Sarlos, S. Chaturvedi
Proceedings of NeurIPS 2024. [pdf]

[21] **Embodied AI with Two Arms: Zero-Shot Learning, Safety, and Modularity**
J. Varley*, S. Singh*, D. Jain*, K. Choromanski*, A. Zeng, S. B. Roy Chowdhury, A. Dubey, V. Sindhwani
Proceedings of IROS 2024 (*Oral*). [pdf]
🏆 **Best RoboCup Paper Award**
🏆 **Nominee for Best Safety, Security, and Rescue Robotics Papers.**

[20] **Incremental Extractive Opinion Summarization Using Cover Trees**
S. Basu Roy Chowdhury, N. Monath, A. Dubey, M. Zaheer, A. McCallum, A. Ahmed, S. Chaturvedi
Proceedings of TMLR 2024. [pdf]

[19] **Enhancing Group Fairness in Online Settings Using Oblique Decision Forests**
S. Basu Roy Chowdhury, N. Monath, A. Beirami, R. Kidambi, A. Dubey, A. Ahmed, S. Chaturvedi
Proceedings of ICLR 2024 (*Spotlight*). [pdf]

[18] **Robust Concept Erasure via Kernelized Rate Distortion Maximization**
S. Basu Roy Chowdhury, N. Monath, A. Dubey, A. Ahmed, S. Chaturvedi
Proceedings of NeurIPS 2023. [pdf]

[17] **Unsupervised Opinion Summarization Using Approximate Geodesics**
S. Basu Roy Chowdhury, N. Monath, A. Dubey, A. Ahmed, S. Chaturvedi
Preliminary version at New Frontiers in Summarization Workshop
Proceedings of EMNLP Findings 2023. [pdf]

[16] **Aspect-Aware Unsupervised Extractive Opinion Summarization**
H. Li, S. Basu Roy Chowdhury, S. Chaturvedi
Proceedings of ACL Findings 2023. [pdf]

[15] **Efficient Graph Field Integrators Meet Point Clouds**
K. Choromanski*, A. Sehanobish*, H. Lin*, Y. Zhao*, E. Berger, A. Pan, T. Parshakova, D. Watkins, T. Zhang, V. Likhosherstov, S. Basu Roy Chowdhury, A. Dubey, D. Jain, T. Sarlos, S. Chaturvedi, A. Weller
Proceedings of ICML 2023. [pdf]

[14] **Sustaining Fairness via Incremental Learning**
S. Basu Roy Chowdhury and S. Chaturvedi
Proceedings of AAAI 2023 (*Oral*). [pdf]

[13] **Learning Fair Representations via Rate-Distortion Maximization**
S. Basu Roy Chowdhury and S. Chaturvedi
Proceedings of TACL (*Oral* at EMNLP 2022). [pdf] [MIT Press]

[12] **Unsupervised Extractive Opinion Summarization Using Sparse Coding**
S. Basu Roy Chowdhury, C. Zhao, S. Chaturvedi
Proceedings of ACL 2022. [pdf]

[11] **Read Top News First: A Document Reordering Approach for Multi-document News Summarization**
C. Zhao*, T. Huang*, S. Basu Roy Chowdhury, M. Chandrasekaran, K. McKeown, S. Chaturvedi
Proceedings of ACL Findings 2022 (*short*). [pdf]

[10] **Adversarial Scrubbing of Demographic Information for Text Classification**
S. Basu Roy Chowdhury, S. Ghosh, Y. Li, J. Oliva, S. Srivastava, S. Chaturvedi
Proceedings of EMNLP 2021. [pdf]

[9] **Is Everything in Order? A Simple Way to Order Sentences**
S. Basu Roy Chowdhury*, F. Brahman*, S. Chaturvedi
Proceedings of EMNLP 2021. [pdf]

[8] **Learning beyond datasets: Knowledge Graph Augmented neural networks for NLP**
K.M. Annervaz*, S. Basu Roy Chowdhury*, A. Dukkipati
Proceedings of NAACL 2018 (*Oral*). [pdf]

Other refereed publications:

[7] **Does Commonsense help in detecting Sarcasm?**
S. Basu Roy Chowdhury and S. Chaturvedi
Proceedings of Insights from Negative Results in NLP workshop, EMNLP 2021. [pdf]

[6] **Towards Efficient Data Normalization Using Grammar-based Parsing**
S. Basu Roy Chowdhury, A. Mishra, A. Banerjee
Proceedings of Microsoft Journal of Applied Research, MSJAR 2020. [pdf]

[5] **Instance-based Inductive Deep Transfer Learning by Cross-Dataset Querying with LSH**
S. Basu Roy Chowdhury, K.M. Annervaz, A. Dukkipati.
Proceedings of DeepLo Workshop, EMNLP 2019. [pdf]

[4] **AdGAP: Advanced Global Average Pooling**
A. Ghosh, B. Bhattacharya*, S. Basu Roy Chowdhury*
Proceedings of Student Abstract and Poster Program, AAAI 2018. [pdf]

[3] **Training Autoencoders in Sparse Domain**
B. Bhattacharya, A. Ghosh*, S. Basu Roy Chowdhury*
Proceedings of Student Abstract and Poster Program, AAAI 2018. [pdf]

[2] **Handwriting Profiling using Generative Adversarial Networks**
A. Ghosh*, B. Bhattacharya*, S. Basu Roy Chowdhury*
Proceedings of Student Abstract and Poster Program, AAAI 2017. [pdf]

[1] **SAD-GAN: Synthetic Autonomous Driving using Generative Adversarial Networks**
A. Ghosh*, B. Bhattacharya*, S. Basu Roy Chowdhury*
Proceedings of DLAI Workshop, NeurIPS 2016. [pdf]

(* = equal contribution)

Patents:

[1] **Fast and Accurate Extractive Summarization**
Vlad Morariu, Somnath Basu Roy Chowdhury, Puneet Mathur, Nikolaos Barmpalios, Ani Nenkova
Assignee: Adobe, Inc. (*Application in Progress*)

Theses:

[2] **Equitable Machine Learning in Static and Dynamic Environments**
Advisor: Prof. Snigdha Chaturvedi
Ph.D. Thesis, UNC Chapel Hill [pdf]

[1] **Lung Cancer segmentation in 3D CBCT Images**
Advisor: Prof. Jayanta Mukhopadhyay
M.Tech. Thesis, IIT Kharagpur [pdf]

Talks

'Concept Manipulation in Deep Learning Models', Jane Street, New York (Apr 2023)
'Learning Fair Representations For Equitable ML', IVLC Seminar, UNC CH (Jan 2023)
'Adversarial Debiasing in NLP', NLP Course, UNC CH (Apr 2022)
'Probabilistically grounded word embeddings', Bing paper reading forum, *Microsoft* (Nov 2019)
'Secure Machine Learning', *Apple India Research Forum* (Apr 2019)

Professional Service/ Teaching

- **Program Committee Member/ Reviewer**

  ICML (2025-), ICLR (2024-), NeurIPS (2023-), AISTATS (2025-), TMLR (2024-), ACL (2022-), ARR (2021-), EMNLP (2021-), LREC-COLING (2024-), NAACL (2024), AAAI (2023-), COLM (2024-), ICRA (2024)

  Narrative Understanding Workshop, NAACL (2021-2022)

  2nd workshop on Insights from Negative Results in NLP, EMNLP (2021)

- *Teaching Assistant*

  (EE21101) Signals and Networks Lab, IIT Kharagpur 2018
  (EE19001) Electrical Technology Lab, IIT Kharagpur 2017

Academic Achievements

- Best RoboCup Paper Award at IROS 2024
- Dissertation Completion Fellowship - Awarded to ∼5 candidates across UNC-CH 2024
- Top Candidate for Jane Street Graduate Research Fellowship 2023, 2024
- Student Volunteer Award, ACL 2022 (*declined*) 2022
- Travel Award from GPSG, UNC Chapel Hill for EMNLP 2021 (*declined*) 2021
- All India Rank 7 (EECS) - Nationwide Education & Scholarship Test (NEST - II). 2017
- ACM-ICPC Asia-Chennai Regionals - Rank 33/1100 2015
- IAS Summer Research Fellowship - National Brain Research Center 2015
- INSPIRE Scholarship - Top 1 percentile in ISC (Grade 12). 2013
- Top ranks - JEE (Mains, Advanced): 99+ percentile, Top 1% in ISCE & ISC 2013

Miscellaneous Activities

- *Winner* of the WaterAid Marathon Hyderabad - 5km category 2019
- *Best Innovation Award* - Apple EMEIA Hackathon 2018
- *Special Mention in Technology GC* - Graduating batch LBS Hall. 2018
- *Silver-medal* winning Data Analytics team - Inter-hostel GC 2017
- *Gold-medal* winning Hardware Modeling team- Inter-hostel GC 2015
- 5th year distinction in *Fine Arts* and Painting - All India Art and Crafts Society 2010